Assignment 1 Report
Digital Signal Processing
Fourier Transform

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October 16, 2026

Introduction

We have developed software to transform, filter, and modulate audio signals utilising Fourier transforms and the frequency domain.

1 Audio Plot

Part 1 contains the logic for obtaining the frequency domain representation of our audio signal, involving reading the WAV file, normalizing the signal, performing a fast Fourier transform, and plotting the graphs in the time and frequency domains.

1.1 Read_WAV_file

Reads and normalizes the data to a range of [-1:1] (32-bit float). If a stereo input is given, only the first channel is stored. Automatically handles 8-bit unsigned, 16-bit signed, and 32-bit signed integer PCM.

Inputs:

Outputs:

1.2 Plot_time_spectrum

Plots a time domain graph representing the audio signal (amplitude against time).

Inputs:

Outputs:

1.3 calculate_fft

Takes a fast Fourier transform of the audio data, producing a full complex spectrum.

Inputs:

Outputs:

1.4 get_nyquist_spectrum

Produces a Nyquist-adjusted frequency spectrum for plotting purposes.

Inputs:

Outputs:

1.5 plot_frequency_spectrum

Plots a frequency domain ṡvg plot in the specified output folder, with optional logarithmic scaling.

Inputs:

Outputs:

2 Audio Enhancer

Frequency filtering and amplification.

2.1 generate_wav_file

Generates a WAV file, designed to process frequency-modulated audio data.

Inputs:

Outputs:

2.2 calculate_ifft

Performs an inverse Fourier transform on the input complex spectrum to produce a reconstructed time-domain data array.

Inputs:

Outputs:

2.3 raised_cosine_filter

Uses a cosine smoothing method to modulate/remove specific frequency ranges without producing ringing from Gibb's phenomenon.

Inputs:

Outputs:

3 Part 3 Aural Exciter

Utilizes the functions from the previous two parts of the assignment to digitally recreate an aural exciter circuit as given in the specification. An inverse tan function is applied to a high-pass filtered portion of the input signal to introduce non-linear harmonic distortion. These newly generated high-frequency harmonics are then scaled and mixed back with the original dry audio, enhancing the signal's brightness and presence.